

Jackson Barber

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Education

University of Waterloo - BAsC in Mechatronics Engineering - 3.92 GPA Sept 2025 – Sept 2030

- Awards: David Gerard Conroy Undergraduate Engineering Scholarship

Skills

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- **Modeling / Simulation:** Solidworks, Fusion360, OnShape, Autocad, 3D Printing, Traditional & Master Modeling, Ansys
 - **Manufacturing:** CNC Mill, Manual Mill, Lathe, Thermoformer, Horizontal & Vertical Bandsaw, Mig & Arc Welding
 - **Electrical / Controls:** Kicad, Altium, Soldering, Crimping, Sensor Use, Stepper and Servo Motor Use

Experience

Automation Engineer – Precision Resource - Cambridge, ON May 2026 - Present

- Designed autonomous inspection allowing for the removal of human QA
- Utilized FANUC robot arms to ensure precision and repeatability
- Designed universal end of arm tooling, removing the need for quick change modules saving up to \$20,000
- Used RobotGuide to assist senior engineers with PLC programming

Mechanical Project Manager – Waterloo Aerial Robotics Group (WARG) - Waterloo, ON Dec 2025 - Present

- Designed and optimized an avionics lid for the competition drone using surface modeling and draft analysis in SolidWorks, reducing weight by 59% while maintaining structural integrity
- Developed and simulated a custom firefighting nozzle using CFD analysis, increasing exit velocity by 53%
- Designed a CNC hotwire cutter for custom wing designs, incorporating an anti-backlash nut to ensure accuracy.
- Modeled and manufactured 3+ electronics enclosures in SolidWorks, ensuring accessible I/O ports, PCB mounting security, and streamlined assembly

Lead Designer – WARP7 Robotics Team - Toronto, ON Sept 2022 - May 2025

- Led strategic design by developing pros-cons analysis templates, match strategy, and data-driven scouting documents
- 3D modeled and manufactured manipulator mechanisms in SolidWorks, drafting technical drawings with GD&T specifications and fabricating components using CNC and manual machining processes
- Developed educational resources and ran technical seminars, achieving a 33% increase in rookie member engagement
- Implemented skeleton sketches and master modeling workflows to enable parametric design iteration

Projects

Custom Mouse Model and PCB Design

- Designed an ergonomic mouse form using clay modeling and photogrammetry mesh generation to prevent repetitive strain injuries during extended use
- Refined the model in Fusion360 using surface modeling, integrating mounting for PCBs, switches, and sensor alignment
- Designed custom PCB in KiCad using a Pro Micro microcontroller and PMW3389 optical sensor, implementing SPI communication protocols for high-precision tracking

Precision Disc Launcher

- Designed and manufactured a precision disc launcher with dual-axis spin control (pitch and yaw), achieving 5cm accuracy at 10m target distance for competitive applications
- Implemented a novel outrunner motor configuration with the motor body directly spinning the launching wheel, reducing overall assembly size and weight by 30% compared to traditional shaft-driven designs
- Manufactured custom aluminum components using CNC and manual mills and lathes, maintaining tolerance on all parts
- Employed type 2 interference fits to secure components while preserving motor electronics

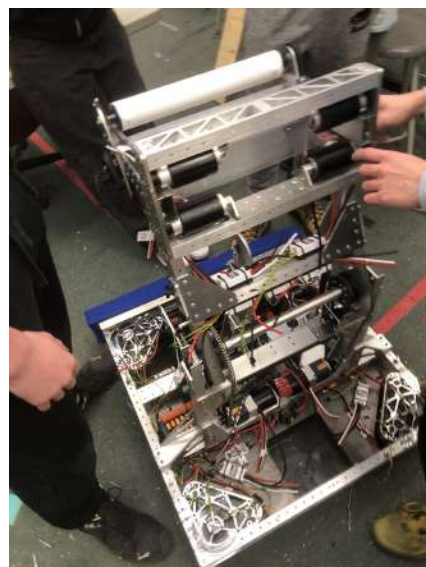
Custom Mouse

- Created a clay outline to fit my hand, used photogrammetry in meshroom to create a mesh which could then be adapted in CAD software
- Adapted the CAD model in Fusion360 using surfaces to create a printable model which could hold the buttons, PCB, and sensor.
- Designed wiring schematics and PCB for mouse function using an Arduino ProMicro and a PMW3389 sensor
- Features left and right click, forward and back, scroll wheel rotation/press, and adjustable DPI
- Connects to any device using a USB-Micro and communicates using the AdvMouse library



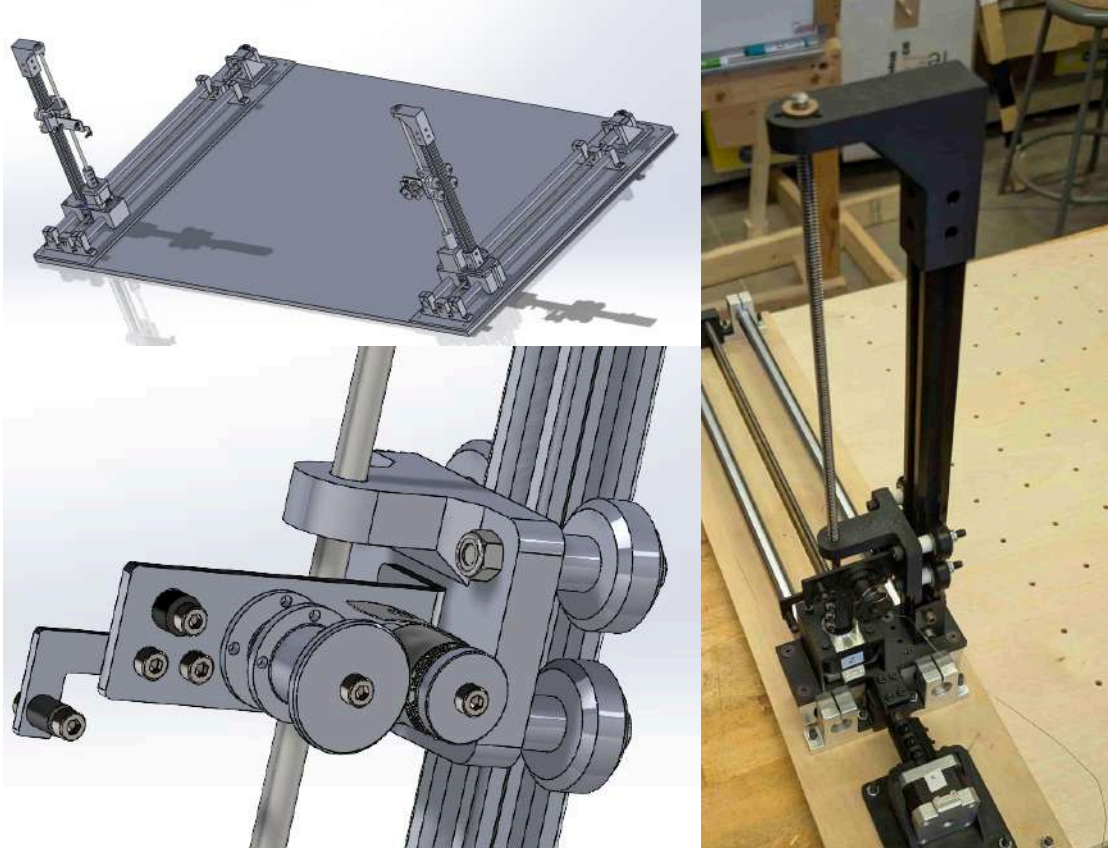
FRC 2024 Disc Launcher @ WARP7

- Personally designed and manufactured a manipulator to shoot foam rings capable of applying spin in both the pitch and yaw axis.
- Used a unique outrunner design having the wheel spin on the motor rather than through a shaft. This design allowed for a 30% reduction in size and 25% reduction in weight when compared to the average team.
- Integrated a hall effect sensor to allow for autonomous shooting sequence
- Designed using type 2 interference fits to ensure strong hold while not damaging the attached motor
- Fabricated the parts using both CNC and manual milling machines



CNC Hotwire Cutter @ WARG

- Designed a CNC hotwire cutter to allow for easier wing design for fixed wing aircraft
- Incorporated an anti-backlash nut using a set screw to prevent slop
- Designed an adjustable limit switch to ensure an accurate homing sequence
- Run through an SKR Pico using a modified version of the GRBL CNC library
- Adapted custom DXF airfoil profiles into Gcode using JediCut



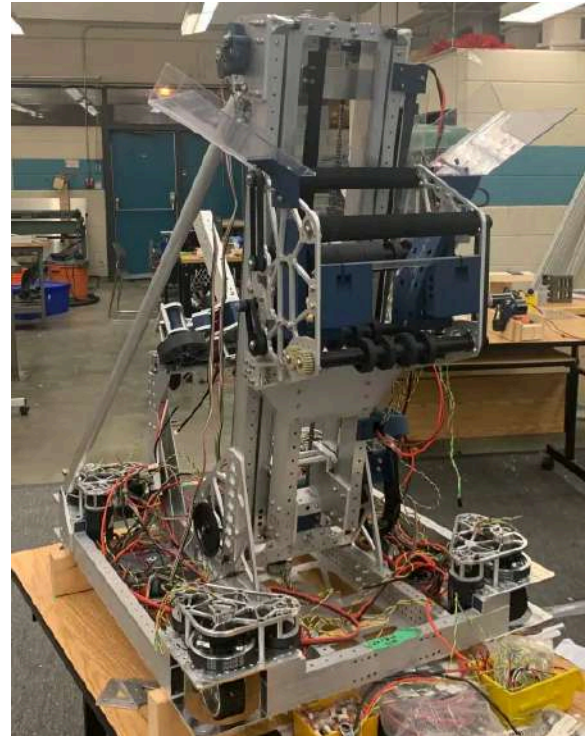
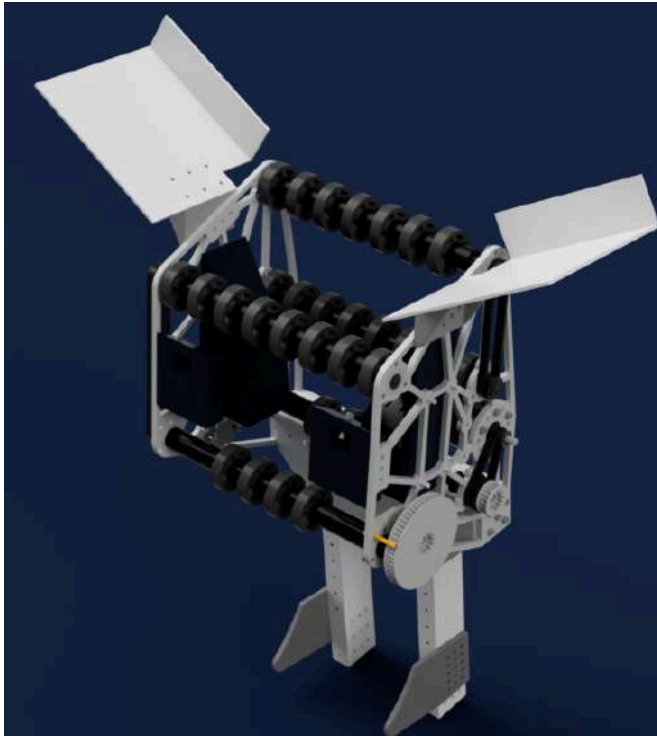
Custom Machined Aluminium Chess Set

- Manually machined a custom aluminium chess set
- Worked off engineering drawings to ensure piece consistency
- Measured pieces using micrometers, calipers, and machine DRO



FRC 2025 PVC Pipe Manipulator @ WARP7

- Personally designed and fabricated the intake/outtake mechanism using a master modelling workflow
- Designed modularly to allow for additional DOF, change of motors, or increased compression
- Designed polycarbonate and 3DP guides to allow for intake at any angle -> eject vertically downward
- Incorporated a second front intake that allowed it to hold and shoot yoga balls into a net
- Created custom pocketing pattern reducing plate weight by 68% while maintaining structural stability



Avionics Lid @ WARG

- Developed the waterproofing lid for the 2026 competition drone using solidworks surfaces and draft analysis
- Manufactured by thermoforming 0.020" PETG on a custom routed XPS mold
- Included a flexure near the mounting point to reduce load and prevent snapping
- Custom clip mounting system allowed for easy mounting
- Added relief holes to ensure strong suction while thermoforming
- Flew at AEAC 2026, ensuring reliability under rainy conditions

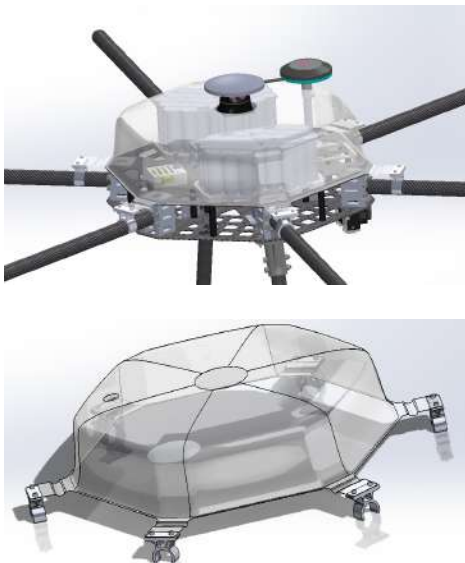
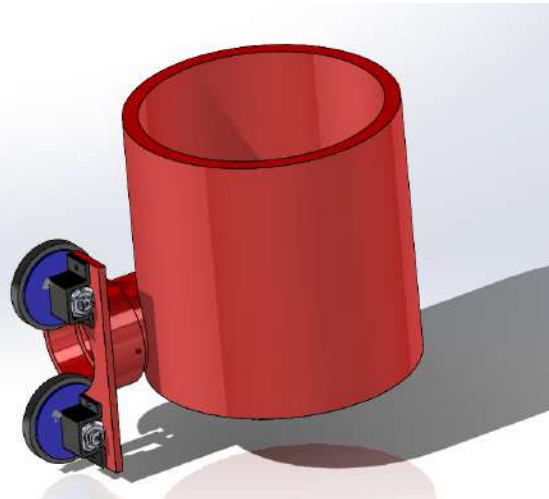
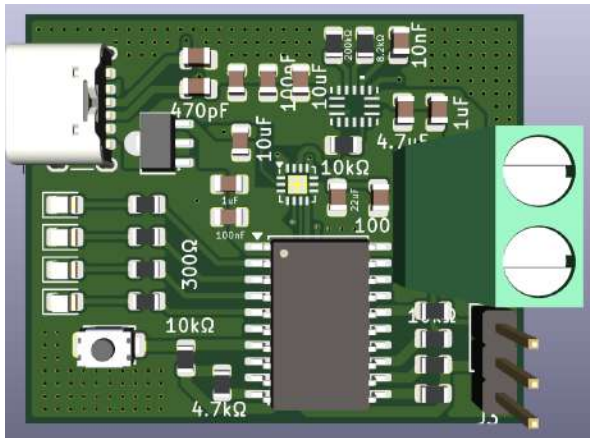
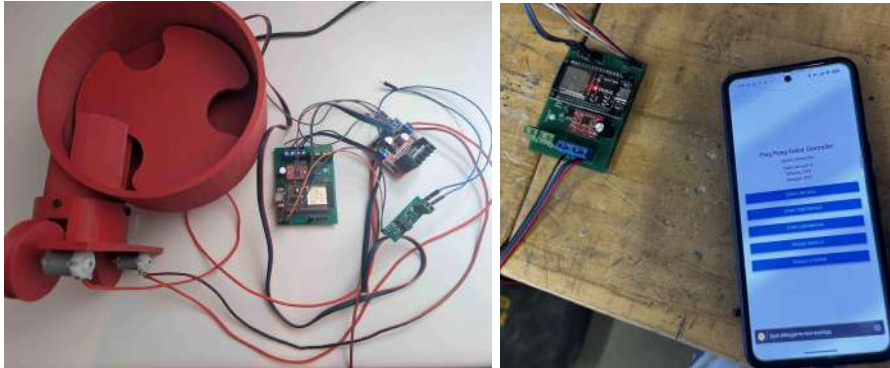


Table Tennis Serving Robot

- Designed a robot capable of serving ping pong balls for serve receive practice
- Runs from a single USB-C connection and is controlled using a custom mobile app
- Designed boards for: USB-C power negotiation, motor and microcontroller breakout, and buck voltage regulation
- Uses a printed spindlexer and wheels to allow for spin control
- Mobile app was designed using Expo for React Native allowing for simple end user bluetooth connection
- Fully modular electronic design allows for reusability of power boards



Custom Smart Watch

- Designed a custom smart watch to allow for more control over software and hardware integration
- Incorporates step counting, music control, touch screen, heart rate sensing, USB-C charging, and notification vibration
- Built around the nRF54L15 to allow for bluetooth connectivity while staying at low power
- Contained in a custom printed case allowing for a 30% lighter design and personal customization
- Projected to start integration testing July 2026

